

Submission PHYS1-QUIZ-SUB06

Question PHYS1-QUIZ-Q01

notation changes halfway:

1. $v \approx v_i + at \approx 3 + 3(2) \approx 9 \text{ m/s}$

2. $\Delta x \approx v_i t + \frac{1}{2}at^2 \approx 3(2) + \frac{1}{2}(3)(2)^2 \approx 12 \text{ m}$

marks, subscripts, and units are mixed up

result written as: $v=9 \text{ m/s}$; displacement=12 m (please interpret)

Question PHYS1-QUIZ-Q02

notation changes halfway:

1. $N \approx mg \approx 68.6 \text{ N}$; friction $\approx \mu N \approx 13.7 \text{ N}$

2. $\Sigma F \approx 27 - 13.7 \approx 13.3 \text{ N}$

3. $a \approx \Sigma F/m \approx 1.90 \text{ m/s}^2$

marks, subscripts, and units are mixed up

result written as: $a=1.90 \text{ m/s}^2$; friction opposes motion (please interpret)

Question PHYS1-QUIZ-Q03

notation changes halfway:

1. $mgh \approx \frac{1}{2}mv^2 \Rightarrow v \approx \sqrt{2gh} \approx \sqrt{2 \cdot 9.8 \cdot 8}$

2. $v \approx 12.52 \text{ m/s}$; mass cancels

marks, subscripts, and units are mixed up

result written as: $v=12.52 \text{ m/s}$ (please interpret)

Question PHYS1-QUIZ-Q04

notation changes halfway:

1. $p_{\text{total, before}} \approx p_{\text{total, after}}$

2. It applies when external impulse on the system is negligible.

marks, subscripts, and units are mixed up

result written as: total momentum before equals after when external impulse is negligible (please interpret)

Question PHYS1-QUIZ-Q05

notation changes halfway:

1. Impulse is area under $F(t)$: $J \approx F\Delta t \approx 6(0.5) \approx 3 \text{ N}\cdot\text{s}$.

2. $\Delta p \approx J$, so momentum changes by $3 \text{ kg}\cdot\text{m/s}$ in the force direction.

marks, subscripts, and units are mixed up

result written as: impulse=3 N·s; momentum changes by $3 \text{ kg}\cdot\text{m/s}$ (please interpret)